D3 - Foundational Mathematical Concepts for Machine Learning

Machine Learning Algorithms

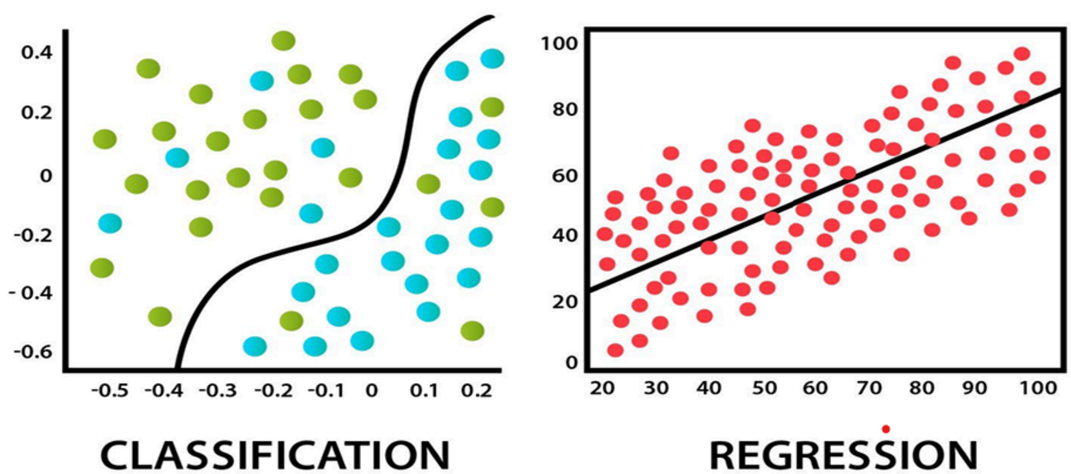
1. Supervised Learning

Supervised learning is a category of machine learning that uses labeled datasets to train algorithms to predict outcomes and recognize patterns. It uses LABELED DATA or GROUND TRUTH.

Broadly classified as:

○ Classification Algorithms (goal is to predict the categorical label (class) of a given input based on training data.)

○ Regression Algorithm (goal is to predict a continuous output variable based on one or more input variables.)



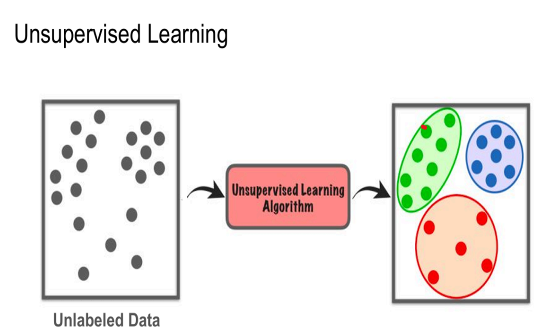
2. Unsupervised Learning

Unsupervised Learning is a type of machine learning that learns from data without human supervision. (UNLABELED DATA)

Common Unsupervised Algorithms:

○ Clustering

○ Dimensionality Reduction



3 . Reinforcement Learning (Reward Penalty System)
4 .Semi Supervised Learning
Vectors

1. vector span
   The span of a set of vectors refers to all the possible vectors you can create by forming linear combinations of them.The span tells you the full set of directions and positions you can reach by combining your vectors with real-number weights.
2. Linearly Independent Vectors
   A set of vectors is linearly independent if none of the vectors in the set can be written as a linear combination of the others. The only way to get the zero vector as a linear combination is to use all zero coefficients.
3. Linearly Dependent Vectors
   A set of vectors is linearly dependent if at least one vector can be written as a linear combination of the others.
   Relation to Span:
   If vectors are linearly independent, each one adds a new direction to the span.

   If they are linearly dependent, at least one vector is unnecessary for spanning the space — it can already be built from the others.

4. Determinant of matrix
   How much a matrix "scales" space when it transforms it.
   It also tells uswhether:
   The matrix is invertible (can be undone).
   The rows or columns are linearly independent.

The **determinant** tells you how the **area** of that square changes:

- `Det = 2` → the area **doubles**
- `Det = 0.5` → the area **shrinks to half**
- `Det = 0` → the square is **squished flat** — it loses a dimension (i.e., no area), which means it can't be reversed!

5. Matrix dotproduct
   The dot product (also called the scalar product) is a way to measure how much two vectors point in the same direction.
   Dot Product and their Meaning

> 0	Vectors point in a similar direction (angle < 90°)
= 0	Vectors are perpendicular (90° angle) → no alignment at all
< 0	Vectors point in opposite directions (angle > 90°)

6. Duality
   "Every vector can be viewed in two ways: as a point or as a function that measures something."

7. Cross Product
   The cross product is an operation between two 3D vectors that gives you a new vector, which:
   - Is perpendicular (90°) to both original vectors.
   - Points in a direction according to the right-hand rule.
   - Has a length (magnitude) equal to the area of the parallelogram formed by the two original vectors.
8. Cramer's Rule
   Cramer's Rule is a method to solve a system of linear equations using determinants. Cramer's Rule uses determinants (geometric quantities) to find the exact values of unknowns in a linear system of equations.
   It only works when:
   You have as many equations as variables
   The determinant of the coefficient matrix is not zero (i.e., the system has a unique solution)
9. Change of Basis
   Changing the basis means Expressing the same vector using a different set of coordinate axes.
   Imagine instead of measuring direction using "right" and "up," you measure using "tilted" directions — a new pair of basis vectors.

10. Eigenvector and Eigenvalues
    When we have a transformation (like stretching, rotating, flipping, etc.) that acts on vectors.

    Most vectors will change direction when transformed.

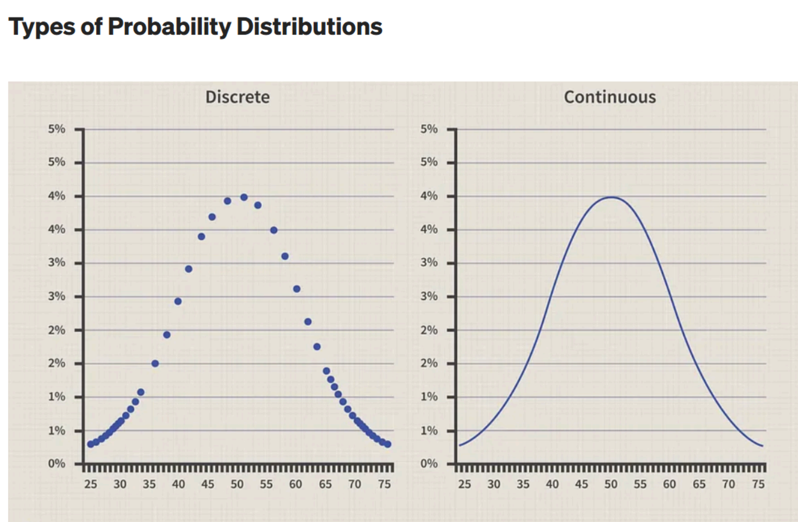
    But some special vectors don't change direction — they only get stretched or squished.

    These special vectors are called eigenvectors.

The amount by which they're stretched or squished is the eigenvalue.

Probability Density Function
A probability density function (PDF) is a function that describes the probability distribution of a continuous random variable. It gives the probability of a particular outcome falling within a specific range of values. For example, the height of people is a continuous random variable, and its PDF would give the probability of a person's height falling within a particular range of values.

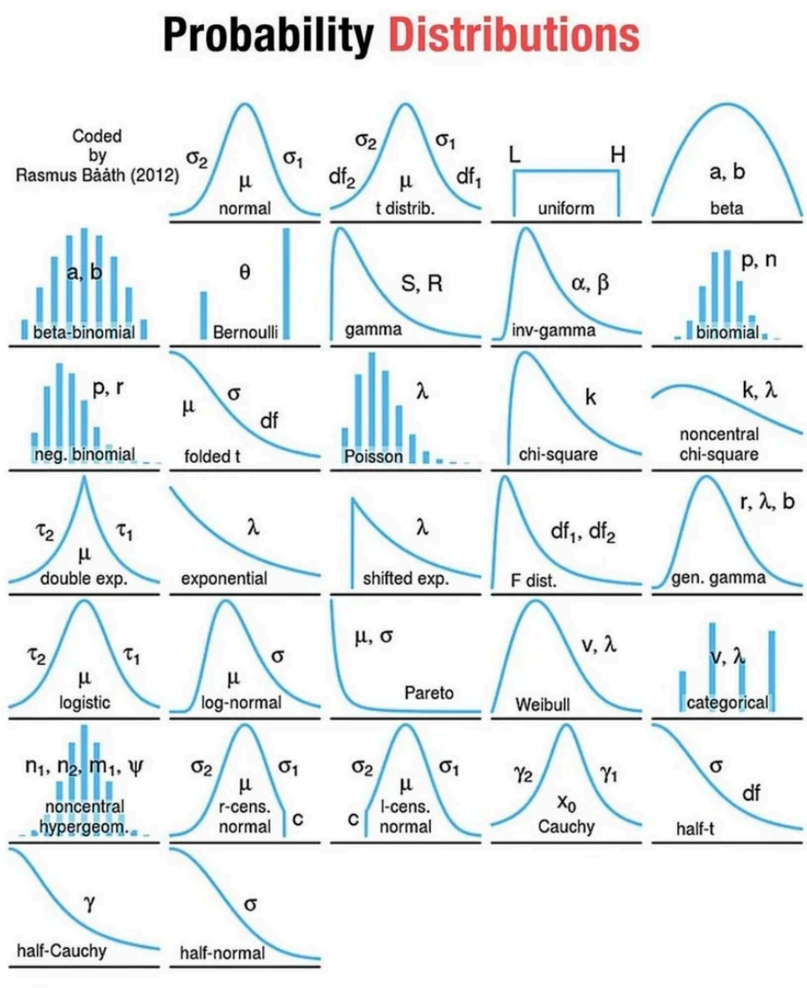
Probability Mass Function
Probability mass function (PMF) is a function that describes the probability distribution of a discrete random variable. It gives the probability of a particular outcome occurring. For example, rolling 10 dice together is a discrete random variable, and its PMF would give the probability of each possible outcome of rolling the dice.



Types of Probability Distributions

A Discrete probability distribution is used when the possible outcomes of a random variable are countable and distinct.
Examples of discrete probability distributions include the binomial distribution, the Poisson distribution, and the geometric distribution.

A Continuous probability distribution, on the other hand, is used when the possible outcomes of a random variable are uncountably infinite and form a continuous range of values.
Examples of continuous probability distributions include the normal distribution, the exponential distribution, and the beta distribution.



When we plot a graph for our data, we usually use the x-axis to represent possible outcomes and the y-axis to represent the probability of those outcomes. Histogram graphs are used for discrete values, while continuous graphs are used for continuous values.

Why are Probability Distributions important?

- Gives an idea about the shape/distribution of the data.

- And if our data follows a famous distribution then we automatically know a lot about the data.

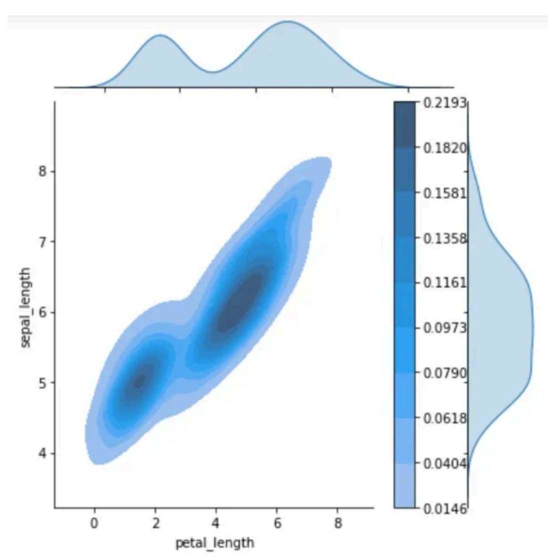
Parameters in probability distributions are numerical values that determine the shape, location, and scale of the distribution. Different probability distributions have different sets of parameters that determine their shape and characteristics, and understanding these parameters is essential in statistical analysis and inference.

- ## Examples of Probability Density Function

1. The normal distribution PDF, which is a bell-shaped curve that is symmetric about the mean and has a fixed variance. It is commonly used to model natural phenomena such as heights, weights, and test scores.
2. The log-normal distribution PDF, which is a skewed distribution that is commonly used to model phenomena such as income and wealth, as well as some natural phenomena such as the sizes of earthquakes and meteorites.
3. The Poisson distribution PDF, which is a discrete distribution that is used to model the probability of a certain number of events occurring in a fixed interval of time or space. It is commonly used in fields such as biology, economics, and physics.

2D Density Plots

A 2D density plot is a graphical representation of the distribution of a two-dimensional dataset that shows the density of points over a 2D space, typically using color-coded contours or heatmaps to indicate areas of high or low density.



2D Density Plots

Normal Distribution

A normal distribution (also called a Gaussian distribution) is a type of continuous probability distribution.It looks like a bell-shaped curve, and it's symmetric around the center.

Key Characteristics:

- Symmetric: The left and right sides of the curve are mirror images.
- Mean = Median = Mode: The peak of the curve is at the center (average value).
- Most values cluster around the mean, and the farther you go from the mean, the less likely the values are.
- About 68% of the data falls within 1 standard deviation of the mean.
- About 95% falls within 2 standard deviations
- About 99.7% falls within 3 standard deviations.

Central Limit Theorem (CLT)

If you take a large number of random samples from any population (doesn't have to be normal), and calculate the average (mean) of each sample, the distribution of those sample means will be approximately normal — no matter what the original population looks like.

Requirements:

- The samples must be random.
- The sample size should be large enough (commonly $n \geq 30$).
- The samples should be independent.

Its Importance

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